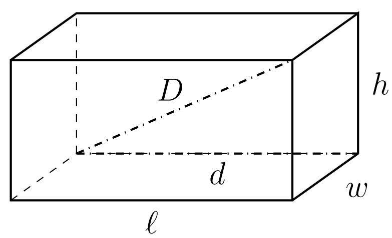
# Multistep Spatial and Coordinate Problems

*Right-triangle sides, coordinate distances, box diagonals, and multistep journeys*

**Directions**

- $a^2+b^2=c^2$, with $c$ the hypotenuse. On a grid, the horizontal and vertical gaps between two points are the legs and the segment joining them is the hypotenuse.
- Inside a box, work in two stages: first the diagonal across the base, then the diagonal of the box using that answer as a leg.
- The problems mix the four methods on purpose. Decide which one a question needs before you calculate.
- Round to two decimal places when the answer is not exact, and write the unit on the answer line.




How every box in this worksheet is labelled: $d$ is the diagonal across the base and $D$ is the diagonal of the box. No values shown — use the numbers given in each problem.

**1.** A right triangle has legs of 8 cm and 15 cm. Find the hypotenuse.

**Answer:** ________________ cm

**2.** Find the distance between the points $(1, 2)$ and $(7, 10)$, in grid units.

Answer: ________________

**3.** A right triangle has a hypotenuse of 20 cm and one leg of 12 cm. Find the other leg.

**Answer:** ________________ cm

**4.** A rectangular field measures 30 m by 40 m. Rosa walks along two sides from one corner to the opposite corner; Ken cuts straight across the diagonal.

(a) How long is the diagonal?

**Answer:** ________________ m

(b) How many metres does Ken save?

**Answer:** ________________ m

**5.** Triangle $PQR$ has vertices $P(0,0)$, $Q(6,0)$ and $R(6,8)$. All lengths are in grid units.

$PQ =$ _________ $QR =$ _________ $PR =$ _________ Perimeter $=$ _________

**6.** A closed box has a base measuring 12 cm by 9 cm and a height of 8 cm.

(a) Find $d$, the diagonal across the base.

**Answer:** _________________ cm

(b) Find $D$, the diagonal of the box.

**Answer:** _________________ cm

**7.** The legs of a right triangle are $x$ and 24, and the hypotenuse is 25. Substituting gives the equation $x^2 + 576 = 625$.

Both solutions: $x =$ ________

Explain which solution is the length of the leg, and why the other one has to be rejected.

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**8.** A pencil box is 20 cm long, 15 cm wide and 9 cm deep.

(a) Find the longest straight line that fits inside the box.

**Answer:** ________________ cm

(b) A pencil is 26 cm long. Explain whether it will fit inside the closed box.

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**9.** A wheelchair ramp rises 1.5 m over a horizontal run of 3.6 m. The ramp, the rise and the run form a right triangle, and edging is fitted along all three sides.

(a) How long is the sloping ramp?

**Answer:** ________________ m

(b) How much edging is needed in total?

**Answer:** ________________ m

**10.** Triangle $ABC$ has $A(-1, 2)$, $B(3, 5)$ and $C(6, 1)$, in grid units.

$AB =$ ________ $BC =$ ________ $AC =$ ________

Show that the triangle is isosceles, using the three lengths you found.

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**11.** A rectangular swimming pool is 25 m long, 10 m wide and 2 m deep everywhere. A rope is stretched from a corner at the water's surface to the opposite corner on the floor of the pool.

(a) Find the diagonal across the surface of the pool.

**Answer:** ________________ m

(b) Find the length of the rope.

**Answer:** ________________ m

**12.** A rectangular garden has corners at $(0,0)$, $(12,0)$, $(12,5)$ and $(0,5)$, measured in metres. A fence runs right around the outside, and one straight support runs from $(0,0)$ to $(12,5)$.

(a) How long is the diagonal support?

**Answer:** ________________ m

(b) How long is the fence around the outside?

**Answer:** ________________ m

(c) How much fencing and support are needed altogether?

**Answer:** ________________ m

(d) Explain why the support is shorter than half the perimeter of the garden.

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